

Bisync AWS Target Architecture

Production-ready design for **1,000 locations** with identical **UAT** and **Prod** environments, including platform AI (onboarding, guided learning, customized reports). Based on the current Bisync.cloud build (ASP.NET Core + React monolith on Cloud Run / Cloud SQL PostgreSQL).

\$13,131 Monthly (UAT + Prod)	\$157,574 Yearly (UAT + Prod)	\$6,566 Per environment / mo	+\$2,268 AI add-on / env / mo
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Currency: USD · Pricing model: On-Demand · Region: Asia Pacific (Singapore) **ap-southeast-1** · Excludes tax and AWS Support. Estimates from AWS Pricing Calculator (Aug 2026).

1. Scope & design principles

This architecture maps the current Bisync production shape to AWS at scale for multi-tenant restaurant operations (location partitions, optional per-company databases, POS offline/LAN sync, HQ/admin SPA).

- **Identical UAT and Prod** — same topology and sizing for this estimate (as requested).
- **Region** — Singapore (ap-southeast-1), closest AWS analogue to current GCP asia-southeast1.
- **App model** — keep the monolith (API + React SPA) as containers on ECS Fargate behind an ALB.
- **Data model** — Aurora PostgreSQL with RDS Proxy to support shared control-plane + per-company DBs.
- **POS** — remains offline-capable at the edge; cloud handles sync, HQ, RMS, and AI assistants.
- **AI** — Amazon Bedrock + RAG knowledge base for onboarding, how-to guidance, and custom reports.

2. Current build → AWS mapping

Current (GCP)	AWS target
Cloud Run monolith (API + React)	ECS Fargate behind Application Load Balancer
Cloud SQL PostgreSQL 16 + multi-DB tenancy	Aurora PostgreSQL (I/O-Optimized) + RDS Proxy
Same-origin SPA delivery	CloudFront → ALB (S3 for static/knowledge assets)
Artifact Registry	Amazon ECR
Secrets / env configuration	AWS Secrets Manager
Cloud Logging / Monitoring	Amazon CloudWatch
Edge protection	AWS WAF on CloudFront / ALB
DNS / TLS	Route 53 + ACM
— (new) Platform AI	Bedrock (Claude + Titan) + OpenSearch Serverless + Lambda

3. Target architecture (per environment)

Each of **UAT** and **Prod** is a full copy of the following stack in a dedicated VPC across 3 Availability Zones.

Layer	Components
Edge	Users / RMS mobile / POS sync → Route 53 → CloudFront + WAF → ACM TLS
Compute	ALB → ECS Fargate (Bisync API + SPA), autoscaling baseline 6 tasks (2 vCPU / 4 GB)
Data	RDS Proxy → Aurora PostgreSQL (writer + reader, db.r6g.2xlarge, 2 TB I/O-Optimized)
AI	App → Lambda/API → Bedrock Claude Sonnet 4.6; RAG via Titan Embeddings + OpenSearch Serverless

Layer	Components
Platform	S3, ECR, Secrets Manager, CloudWatch, NAT Gateway x2, AWS Backup, IAM
Delivery	GitHub Actions → build/push ECR → deploy ECS (mirrors today's GHA → Cloud Run flow)

3.1 Sizing assumptions (1,000 locations)

- Always-on **6 × Fargate tasks** (2 vCPU / 4 GB); headroom to ~12 at peak.
- Aurora **2 nodes** db.r6g.2xlarge (writer + reader), **2 TB** storage, **500 GB** backup.
- API load modeled near **~80 RPS** average for ALB / WAF / CloudFront.
- No Redis in current build — ElastiCache omitted (optional later).
- AI: **~58,000 requests/month** per env (~2 RPM × 16 hrs/day), blended **4,500 in / 1,000 out** tokens, 30% prompt cache.

4. AWS services required

Core platform (both envs)

VPC, NAT Gateway, Application Load Balancer, ECS Fargate, ECR, Aurora PostgreSQL, RDS Proxy, S3, CloudFront, Route 53, ACM, Secrets Manager, CloudWatch, WAF, IAM, AWS Backup.

Platform AI (both envs)

Amazon Bedrock (Anthropic Claude Sonnet 4.6 for generation; Amazon Titan Text Embeddings V2 for RAG), OpenSearch Serverless (vector knowledge base), AWS Lambda (AI orchestration).

Optional later

ElastiCache Redis, SQS/EventBridge for sync pipelines, GuardDuty, AWS Support Business/Enterprise, Reserved Instances / Savings Plans for Aurora and Fargate.

5. Cost estimate (monthly & yearly)

All figures are **USD On-Demand**, exclude tax and AWS Support. UAT mirrors Prod 1:1 for this estimate.

5.1 Totals

Scope	Without AI	With AI	AI add-on
Prod	\$4,298	\$6,566	+\$2,268
UAT (same setup)	\$4,298	\$6,566	+\$2,268
UAT + Prod monthly	\$8,596	\$13,131	+\$4,535
UAT + Prod yearly (×12)	~\$103,151	~\$157,574	+~\$54,423

5.2 Per-environment breakdown (with AI) — ~\$6,566 / month

Service	Role	~/mo USD
Aurora PostgreSQL	Writer + reader, 2 TB, backup, RDS Proxy	\$3,096
Amazon Bedrock (Claude Sonnet 4.6)	Onboarding, how-to, custom reports	\$1,408
OpenSearch Serverless	Vector KB (2 index + 2 search OCU, 50 GB)	\$828
ECS Fargate	6 × 2 vCPU / 4 GB always-on	\$546
CloudFront	CDN / edge (~1.5 TB AP egress)	\$246
CloudWatch	Metrics, logs, dashboards, alarms	\$149
NAT Gateway (×2)	Private subnet egress	\$133
S3	Static + knowledge corpus (~1.2 TB)	\$53
WAF	Web ACL + managed rules	\$47
ALB	Public load balancing	\$34
Secrets Manager	App/DB/AI credentials	\$13
Route 53	Hosted zones + health checks	\$7
ECR	Container images	\$5
Bedrock Titan Embeddings	RAG embeddings	~\$1
Lambda	AI orchestration	~\$0*

* Lambda typically covered by free tier at this invocation volume.

5.3 AI usage model (per environment)

Assumes steady-state AI adoption across 1,000 locations for three product surfaces: onboarding assistants, in-app learning / “how do I...”, and natural-language customized reports.

Workload	Approx. requests / mo	Token profile
Help / onboarding	~40,000	~4,000 in / 800 out
Guided learning	~10,000	~4,000 in / 800 out
Customized reports	~8,000	~8,000 in / 2,500 out
Blended (modeled)	~58,000	~4,500 in / 1,000 out

Cost levers: Mixing Claude Haiku for simple help with Sonnet for reports can reduce Bedrock spend ~30–50%. In practice UAT token volume is often 10–25% of Prod even when infrastructure is identical — this estimate keeps UAT tokens equal to Prod for conservative budgeting.

6. AWS Pricing Calculator reference

Shareable estimate (UAT + Prod, core platform + AI). Open the link and click **Update estimate** if totals are still loading. Share links typically remain valid for about one year.

Estimate name: Bisync 1000 loc UAT+Prod with AI ap-southeast-1

<https://calculator.aws/#/estimate?id=ed93e08b12ece30328c60d0d20a6a165baf25952>

Prior estimate without AI (for comparison): <https://calculator.aws/#/estimate?id=94a4b12ba7dc1d07af753049e6413bda251a9b1a>

7. Exclusions & next steps

- AWS Support (Business/Enterprise typically adds a percentage of monthly spend).
- Taxes, Marketplace fees, and unusual cross-region data transfer.
- Reserved Instances / Compute Savings Plans (can reduce Aurora/Fargate 30–50%).
- Multi-region DR / active-active failover.
- Per-location edge hardware and POS devices (out of AWS scope).

Recommended next planning steps: (1) validate AI request volumes with a pilot cohort, (2) decide Haiku/Sonnet routing by use case, (3) apply 1-year Savings Plans to Aurora + Fargate for budget lock-in, (4) optionally shrink UAT AI token budget while keeping identical infrastructure.

Disclaimer: AWS Pricing Calculator provides an estimate of AWS fees and does not include taxes. Actual fees depend on real usage, selected regions, discounts, and account terms.